P5: multi-seed panel of the P4 token-share mix after fresh TL parents (#307836)

Author(s)
This pre-registration is currently anonymous to enable blind peer-review.
It has one author.

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1) Have any data been collected for this study already?
It's complicated. We have already collected some data but explain in Question 8 why readers may consider this a valid pre-registration nevertheless.

2) What's the main question being asked or hypothesis being tested in this study?
P5 is designed after released P4 findings (AsPredicted #307591; Gate X 2026-08-21; both co-primary criteria observed in one initialization). It is a post-P4, prospectively preregistered multi-seed panel of the P4 token-share apparatus. It does not amend AsPredicted #306780, #306935, #307342, or #307591. It is not an independent confirmation of P4. It is not a population estimate, confidence interval, or inferential test.

For each unused parent-init seed s in {1,2,3}, the P5-owned parent launch wrapper sets torch.manual_seed(s) and torch.cuda.manual_seed(s) immediately before GPT.init_weights() for d8 and d20; constructs the official GPT model only after that Torch CPU/CUDA RNG state is set, or reinitializes every trainable GPT parameter through the pinned GPT.init_weights() call after that state is set; records wrapper SHA and SHA-256 of the serialized initial model state before parent training. Python/NumPy wrapper RNG uses fixed seed 424242. Train a fresh Tagalog WikiText-TL-39 nanochat parent (never load P1.1/P2/P3/P4 weights). If P0-T PASSes, freeze d20 as C0_s and continue it on C1 extra Tagalog, C2 pure English, and C3 the P4-frozen EN/TL mix with q_TL=0.50 Tagalog source-content tokens (reuse P4 packed stream by SHA; quotas 9633792/9633792; interleave seed 42). Does C3_s yield lower held-out Tagalog BPB than C2_s and lower held-out English BPB than C1_s?

R_TL(s) = TL val_bpb_full(C3_s) - TL val_bpb_full(C2_s). Success: R_TL(s) <= -0.01.
A_EN(s) = EN val_bpb_full(C3_s) - EN val_bpb_full(C1_s). Success: A_EN(s) <= -0.01.
Equality at -0.01 counts. Both required for that seed's two-component trade-off pattern. No composite.

The panel result is the count of eligible seeds in both / only-R / only-A / neither, plus ineligible if P0-T BLOCKED. P4 seed 0 is historical disclosure, not a P5 confirmatory cell. C3 is not P3 B3. No early stop. No replacement seed. Run seeds 1 then 2 then 3 through every filed seed. One panel unblinding after every seed's V or ineligible-parent stop. Not CORE/chat/SFT.

3) Describe the key dependent variable(s) specifying how they will be measured.
Primary DVs: official val_bpb_full as in P4. Tagalog val on frozen P1.1 val SHA256 4d51644b84d05050bfc8c515079e60f6e437082b6cce2122e9ed00e7b1db2b1c; English val on WT103-raw val SHA256 874dec29844b3d46fc39e5479ee2dc4b3ba37309d9baf3bba4b5654697f3ae3b. Same carry-forward tokenizer on every arm. BPB = mean token NLL / (ln(2) x mean UTF-8 bytes per token), scripts/p5/evaluate_bpb.py, T=2048, BOS-bestfit, full val, CUDA. meta.val_bpb is not a DV.

Per eligible seed: six child cells C1/C2/C3 x {TL, EN}; contrasts R_TL(s) and A_EN(s) as in Q2. C0_s English val once at U_s, descriptive. P0-T covers C0_s Tagalog (CUDA-only status).

After that seed's val seal (test_access[s]=0), one C3-only secondary test event on EN test SHA256 2bccabc020cbb8d09273cccdc42ed926957b83824ca767c96fb588041b8d434e and TL test SHA256 3bd193458f4c494d84dae345548c0c01cb6cd7275e98d6ed39a41d517a093baf. C1/C2 never tested. Not a test-set R_TL/A_EN. Do not reuse P4 Gate V numbers. P5 protected-secondary policy is one C3-only test event per eligible predeclared seed, hence zero to three total P5 test events; it is not one total test event for the whole three-seed panel. Each test is post-seal, secondary, descriptive, and excluded from seed classification and the panel count table. Before panel Gate X, P0-T/U/V scalar files, evaluator stdout/stderr, and test reports are lockboxed and not inspected; test_access[s] is a count only.

4) How many and which conditions will participants be assigned to?
No human participants. Pin 92d63d4; base_train -> scripts/p5/evaluate_bpb.py; T=2048; B=65536; CORE off. Host contract: NVIDIA A40 48 GB, CUDA 12.8, torch 2.9.1+cu128, container runpod/pytorch:1.0.3-cu1281-torch291-ubuntu2404. Use scripts/p5/env.sh and scripts/p5/env.cuda.sh only; never source scripts/p4/* or scripts/p1/p2/p3/*. Never load P1.1/P2/P3/P4 .pt.

Shared once: carry-forward tokenizer.pkl SHA256 04436b854e0841025a3dd2b46baaeeea07a7ccc252e9f99a19171306f00bc5a8 and token_bytes.pt SHA256 a5dbc1c88f6292696108263072d77115718cc2d8357f7ad4859adfa517cc2132; P4 C3 mix-manifest SHA256 f203c615266bc8c33c358c1de397715791cae33536a9743c8a6bf8cd543cb107; one d4 smoke (seed 0; nonconfirmatory).

Panel K=3. Parent-init seeds 1, then 2, then 3 (same s at d8 and d20 via F5-22 wrapper). Per seed: parent on P1.1 Tagalog train SHA256 2b0474c5700dc1eba14def572aa23cc227e4c59c10c2de3ce6b7bda75d137687, N=294; C0_s = freeze d20 after P0-T PASS. Children D_phase2 = 294 x 65536 = 19267584, fresh Muon+AdamW (load_optimizer=False), peak LR = 0.3 x parent peak LR, warmup 14, terminal ckpt only: C1 extra TL, C2 WT103-raw train, C3 P4-frozen mix (do not reshuffle). For each C3 run, trainer-consumed packed train shard SHA256 values
249e2c5e9d06bf17fe14e03c02e622c9e68d90ba337e1e6e33c237fc723252f5,
d24fe7f933abeb38b277b099385518718d2d57e330e5f9b5fd8b1a534e43444e,
a56a729a0e3c7fd2d1e2e99236a4225bf9a86e55d6d97153063de9d6455fa523,

4adbf8f9afcc9870f46f6298ac22f3691ec073d2f452f73aa65322e3ff6331de and language_origin_mask_sha256 140e174a427a7ddf2126553c53352ec049f72fbed475e2404cd4ef122b309c46 must match P4; matching source JSONLs or manifest alone is insufficient. Within seed: R then S then T, then U, then one C3-only V. P0-T untrained floor uses the same seed s as the parent. P4 seed 0 is not a P5 cell.

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.

Run predeclared seed order 1, then 2, then 3. Continue through every filed seed despite any safe operational status from preceding seeds.

Per seed: P0-T at d8 and d20 before any child token (CUDA-only status). Permitted safe output only: PASS, BLOCKED, or TECHNICAL_STOP. If P0-T BLOCKED: that seed is ineligible; skip its children; do not replace it; remaining filed seeds still run.

If PASS: six-cell val seal then R_TL(s) and A_EN(s) as in Q2. Per-seed grammar: both / only-R / only-A / neither. Panel primary: count table over eligible seeds (plus ineligible). No mean, no CI, no p, no random-effects model, no seed dropping, no extra seed, no "P5 confirms P4." |delta| < 0.01 is not a ranking. Do not retune q_TL/D/delta from P4 magnitudes. C0 English val and C3-only tests are descriptive. Exactly one panel Gate X only after each seed has completed V or reached the filed P0-T ineligible stop. No seed-level unblinding. Up to three C3-only test events, one per eligible seed (zero to three total; not one test for the whole panel).

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

Official JSONLs are byte-identical P4 copies (TL train SHA256 2b0474c5...; EN train SHA256 09ae691c...; val/test as in Q3). Hash mismatch on shared splits = panel-level protocol stop. No reclean. Do not drop documents to chase byte share. Rebuild C3 only if every SHA matches P4.

Panel-level protocol stop (no remaining seed under compromised shared registry; no P5 result label): shared-input breach involving splits, tokenizer, evaluator, packed C3 trainer inputs or language-origin mask, test isolation, forbidden weight lineage, val/test in train, inexact C3 quota/order, new tokenizer, ratio -1, HF Trainer, writing into p1/p2/p3/p4 Hub, MPS/CPU for official GPU gates, child used as parent, NaN/Inf, ClimbMix, missing terminal ckpt.

Seed-level protocol stop: breach demonstrably isolated to an unexecuted seed-specific staging/output path; no result for that seed; no replacement seed; remaining seeds continue only if the shared registry is independently re-verified unchanged.

Technical quarantine/restart: pre-outcome failure with no official terminal checkpoint and no accessed P5 scalar; clean restart from immutable C0_s, fresh optimizer, new empty output dir; log incident.

Never exclude a finite official terminal result for outcome. Do not drop an arm or a seed because a contrast looks bad or because the first two seeds recurred. d4 smoke is not confirmatory.

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.

Frozen P1.1 TL-39 train and official WT103-raw train, no added docs. K=3 unused parent-init seeds {1,2,3}, named before filing. Per seed: parent d8 and d20, N=294; if eligible, three d20 children, N=294, D_phase2=19267584; six confirmatory val cells; one C3-only test event. No fourth seed. No early stop. Sample size is the filed panel, not a stopping rule.

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)

Q1 is "It's complicated": public corpora and P1.1/P2/P3/P4 outcomes exist and were known at design (P4 #307591 both-observed on seed 0). No P5 parent, children, val_bpb_full, or test_bpb has been computed. This PDF locks those unobserved P5 quantities. New study; does not amend #306780, #306935, #307342, or #307591. P5 uses archived public text but prospectively trains new computational models and generates new P5 evaluation outcomes; there are no human participants.

P5 is the separately preregistered multi-seed follow-up identified by P4's one-seed limitation. P1.1 supplies the protected Tagalog corpus/split and BPB measurement apparatus; P2 supplies the canonical WikiText-103-raw stream provenance and matched-branch lineage; P3 supplies the fresh-Tagalog-parent, P0-T, common-parent, and protected-evaluation architecture; P4 supplies the distinct token-share-locked C3 treatment, contrasts, and C3-only test policy. P5 is the first predeclared multi-new-seed recurrence panel for that treatment. P5 changes only the predeclared parent-initialization seed and newly trained descendants. It does not load earlier weights, count P4 seed 0 as a P5 cell, or treat P4's released magnitudes as calibration targets. P5 is computationally independent of P4 in its parent weights, children, ledgers, and P5 outcomes; it is not outcome-independent of P4, because P4's released seed-0 results informed this pre-filed replication panel.

C3 is not P3 B3. Lockbox release only at one panel Gate X. Exploratory only (not P5): byte-balanced mix (P6-B), ratio sweep, replay, FilBench, EWC, CORE, SFT. Gate-plan SHA256 d51115aade9c0b1fb8698eaa33540db2d75b2b27765aaaad1bf14b13b0132092. Pre-filing addendum SHA256 839fcaa3dd6e94bd9546df4880a5892851a54ea31743cb0359adc8faebbe9258. Pin 92d63d4e8bb4df75c3b71618f31ddde2378b2bcd. Mix-manifest SHA256 f203c615266bc8c33c358c1de397715791cae33536a9743c8a6bf8cd543cb107. Addendum fixes wrapper boundary, protocol-stop scope, C3 trainer-input hashes, and panel test policy. Test files stay outside train dirs.

BUNDLE

This pre-registration is part of a bundle which includes:

#307,591 - https://aspredicted.org/if84km.pdf - Title: 'P4: token-share mix after fresh TL parent (nanochat, TL-39/WikiText-103)' #306,780 - https://aspredicted.org/6r6v4v.pdf - Title: 'NANOCHAT-FILIPINO P1.1 - WikiText-TL-39 fixed-budget depth vs held-out BPB' #307,342 - https://aspredicted.org/wd2pc8.pdf - Title: 'P3: TL retention after EN continuation (nanochat, TL-39 then WikiText-103)' #306,935 - https://aspredicted.org/xa56bs.pdf - Title: 'P2: EN retention after TL continuation (nanochat, WikiText-103 then TL-39)'